Basic Principles of Medicine 1
Module: Foundations to Medicine

DLA Title:

Genetic Code and Translation

DLA Videos_Parts 1 and 2

Name of Lecturer: Dr Felicia Ikolo

Email: fikolo@sgu.edu

Objectives: DLA: Genetic code and Translation

- Before lecture ‘Translation and Post-translational Modification’

SOM.MK.I.BPM1.1.FTM.3.BCHM.0064	Explain why the genetic code is collinear with DNA, triplet in nature, redundant (degenerate), and non-overlapping.
SOM.MK.I.BPM1.1.FTM.3.BCHM.0065	Define the terms: reading frame and frame-shift mutation.
SOM.MK.I.BPM1.1.FTM.3.BCHM.0066	Interpret the standard genetic code table. Identify the initiation codon and the three termination codons.
SOM.MK.I.BPM1.1.FTM.3.BCHM.0067	Explain the universal nature of the genetic code (with minor differences in mitochondria) and predict the sequence of a peptide if the coding strand of the DNA is provided.
SOM.MK.I.BPM1.1.FTM.3.BCHM.0068	Discuss how protein synthesis proceeds from the N-terminal to their C-terminal ends and explain how this creates a naming convention for peptides.
SOM.MK.I.BPM1.1.FTM.3.BCHM.0050	List and describe the major function of the following types of RNA in cells: mRNA, tRNA, rRNA, snRNA, miRNA.
SOM.MK.I.BPM1.1.FTM.3.BCHM.0070	Describe the subunit composition prokaryotic (70S) and eukaryotic (80S) ribosomes. Recognize the roles of A, P, and E sites on the ribosome.
SOM.MK.I.BPM1.1.FTM.3.BCHM.0071	Describe the structure, function, and charging of tRNAs.
SOM.MK.I.BPM1.1.FTM.3.BCHM.0072	Describe the codon/anticodon interaction and discuss the wobble hypothesis.

Recommended Reading

- **Lippincott's Biochemistry**: Chapter 32

  Pages 496 - 511

Protein Synthesis | Lippincott® Illustrated Reviews: Biochemistry, 8e | Medical Education | Health Library (lwwhealthlibrary.com)

  - **Concept map:** page 512
  - **Questions:** 32.1-32.3, 32.5-32.9

Additional Resources

YouTube videos:

- The big picture of translation:
  - http://www.youtube.com/watch?v=5bLEDd-PSTQ
    - 3 minutes 32 seconds.
- Interpreting the genetic code:
  - https://www.youtube.com/watch?v=-Ht81lHiJac
    - 4 minutes 16 seconds.

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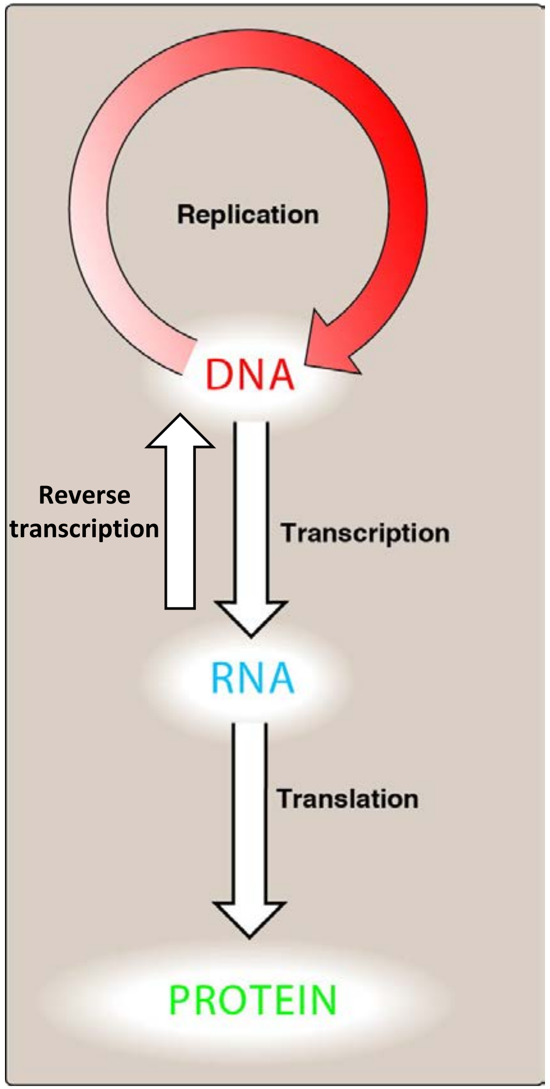
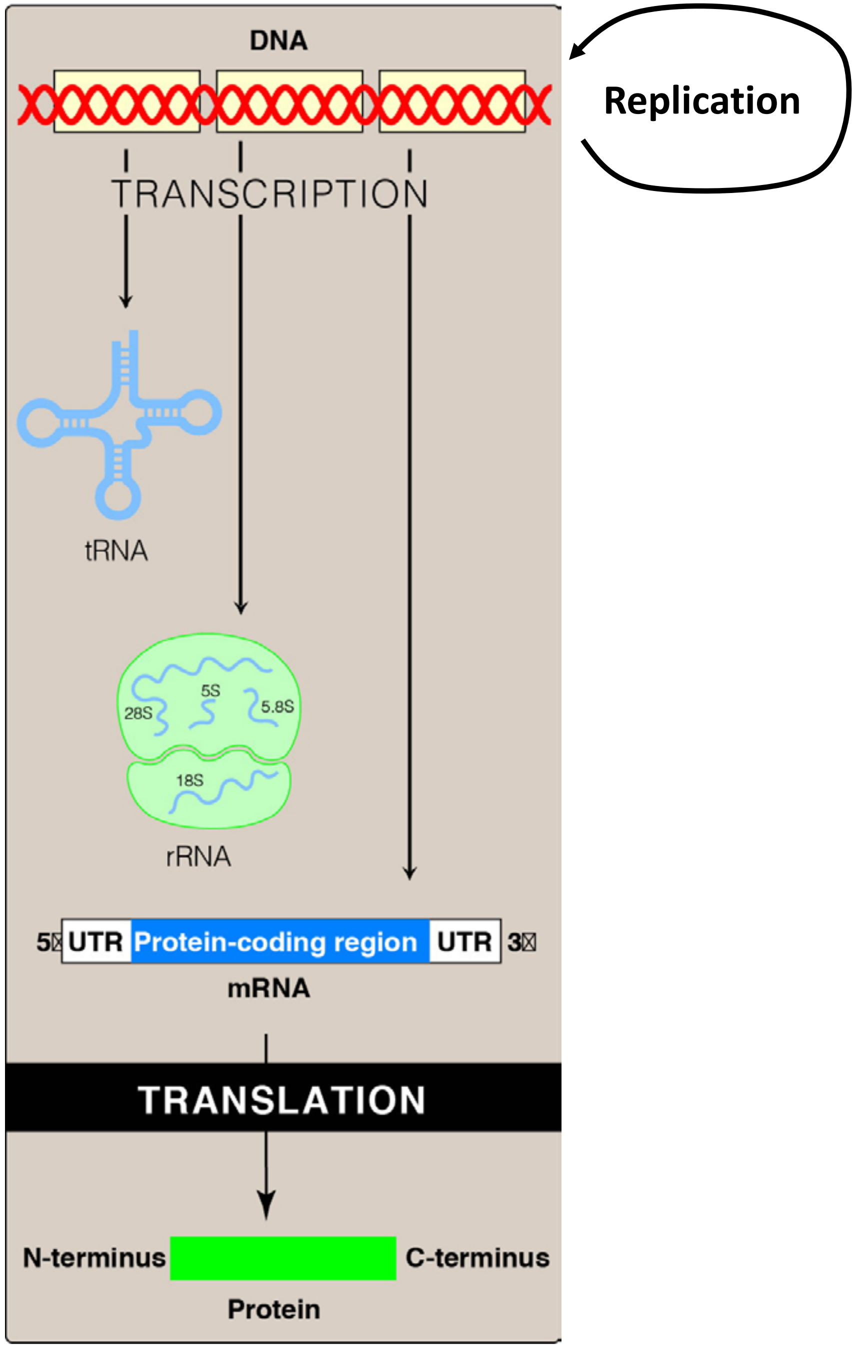
Central Dogma of Genetics and Molecular Biology



Adapted from Lippincott's: page 459



- DNA directs the synthesis and the orientation of RNA

DNA is written 5’ → 3’

RNA is written 5’ → 3’

- RNA directs the synthesis and the orientation of a polypeptide

Protein is written from the N-terminus to the C-terminus

Naming Convention of Peptides

- Peptides are synthesized and named from the N-terminus to the C-terminus, going from left to right.

- Name this tripeptide: Pro-Arg-Gly
  - Prolyl-Arginyl-Glycine

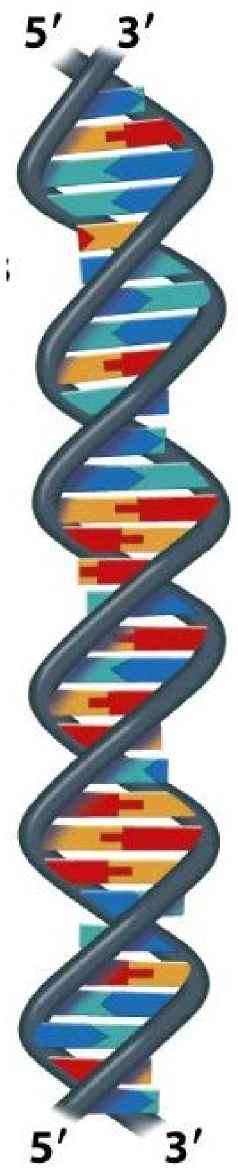
The codon is 3 nucleotides

- How can 4 bases specify the twenty different amino acids?

- If…1 base coded for an amino acid
  - Then 4^1 = 4 amino acids

- If…2 bases coded for an amino acid
  - Then 4^2 = 16 amino acids

- If…3 bases coded for an amino acid
  - Then 4^3 = 64 possible amino acids
  - **Some amino acids must have more than 1 codon**
  - **The genetic code is redundant (degenerate)**



The Genetic Code

- 3 nucleotides make up the codon which specifies an amino acid in a polypeptide chain
- All 64 codons were deciphered by the mid-1960s
- Of the 64 triplets, 61 code for amino acids
- 1 triplet is a start codon

5'-BASE	U	C	A	G	3'-BASE
	MIDDLE BASE				
U	Phe	Ser	Tyr	Cys	U
	Phe	Ser	Tyr	Cys	C
	Leu	Ser	Stop	Stop	A
	Leu	Ser	Stop	Trp	G
C	Leu	Pro	His	Arg	U
	Leu	Pro	His	Arg	C
	Leu	Pro	Gln	Arg	A
	Leu	Pro	Gln	Arg	G
A	Ile	Thr	Asn	Ser	U
	Ile	Thr	Asn	Ser	C
	Ile	Thr	Lys	Arg	A
	Met	Thr	Lys	Arg	G
G	Val	Ala	Asp	Gly	U
	Val	Ala	Asp	Gly	C
	Val	Ala	Glu	Gly	A
	Val	Ala	Glu	Gly	G

1 These four rows show 16 amino acids whose codons begin (5') with A.

2 This column shows 16 amino acids whose codons have the middle base U.

3 These four, separated rows show 16 amino acids whose codons end (3') with G.

4 The codon 5'-AUG-3' designates methionine (Met).

- 3 triplets are "stop" signals to end translation

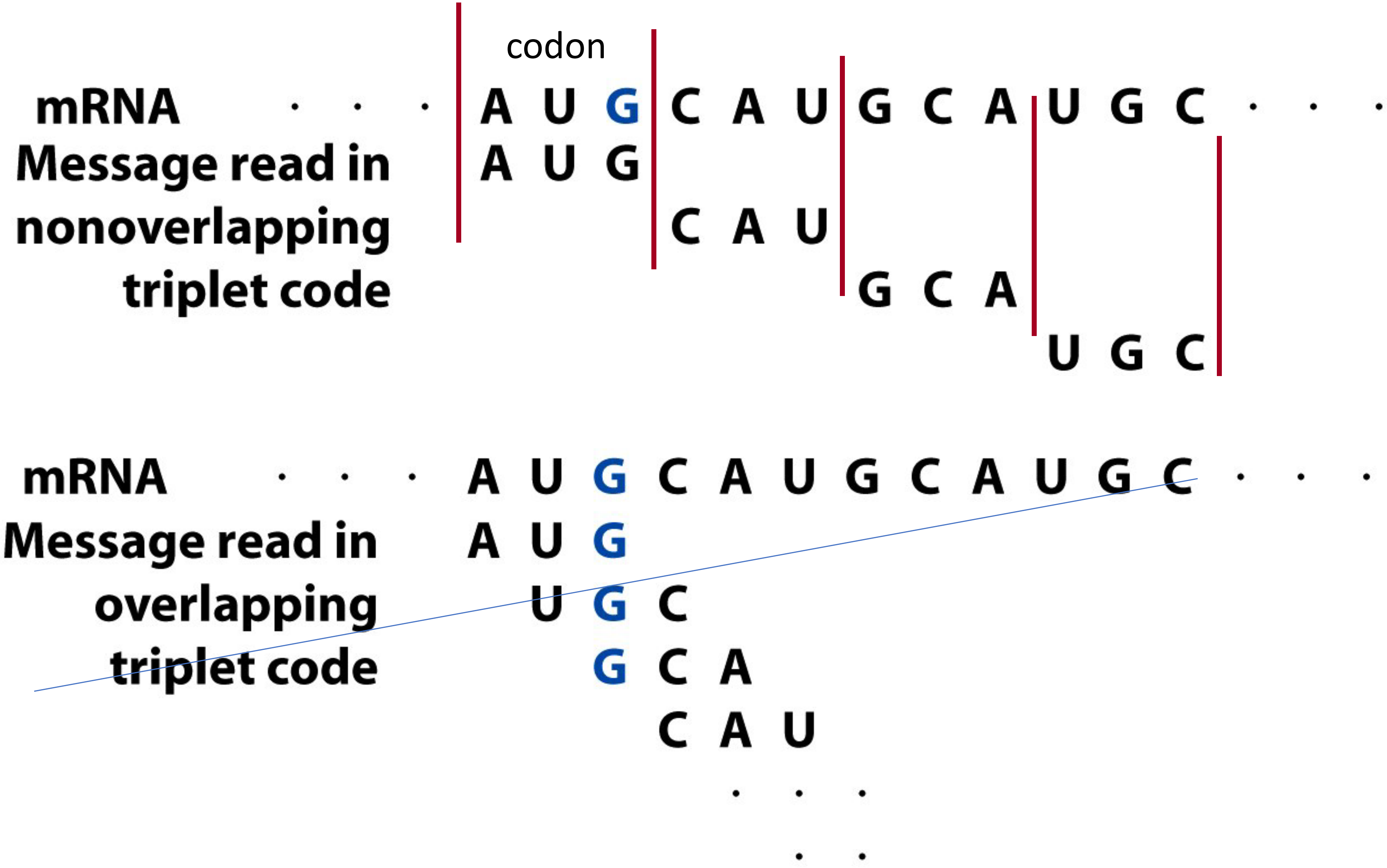
UAA
yo<u>U</u> <u>A</u>re <u>A</u>way

UGA
yo<u>U</u> <u>G</u>o <u>A</u>way

UAG
yo<u>U</u> <u>A</u>re <u>G</u>one

Met and Trp are the only aa that have only one codon

The genetic code is non-overlapping…



… and without punctuation

The genetic code is non-overlapping and comma free

Normal mRNA: AUG ACA CAU AAC GGC UUC GUA

translation

Amino acids: Met Thr His Asn Gly Phe Val

You should be able to predict the sequence of a protein if the RNA sequence is provided.

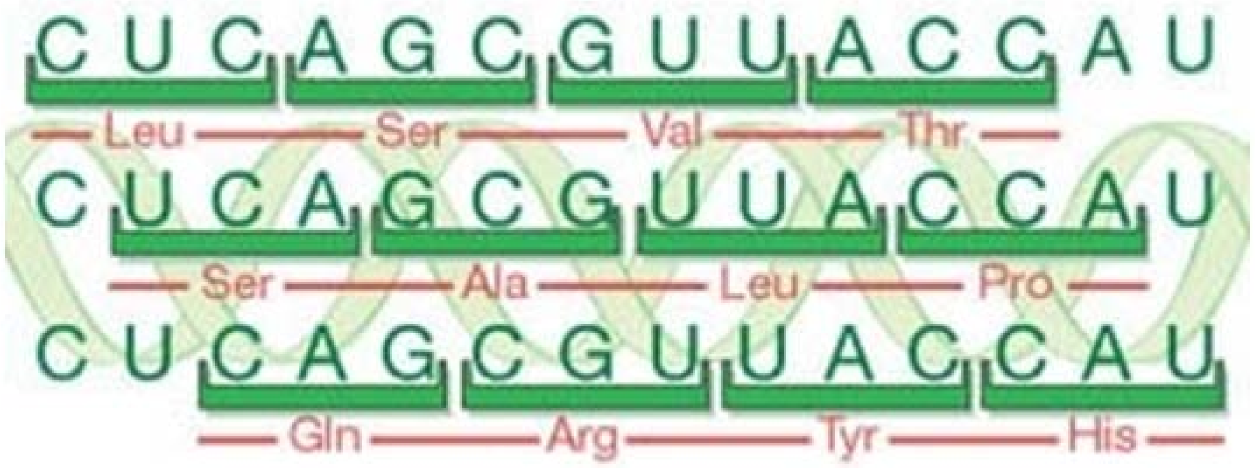
The Genetic Code is Nearly Universal

(a)Tobacco plant expressing a firefly gene

(b) Pig expressing a jellyfish gene

- Shared by the simplest bacteria to the most complex animals
- Genes can be transcribed and translated after being transplanted from one species to another
- Suggests that the genetic code has been conserved from the early stages of evolution
- Minor differences do exist (mitochondria)
  - For instance, in mitochondria, trp has two codons
  - Main Point: Minimize the impact of the change

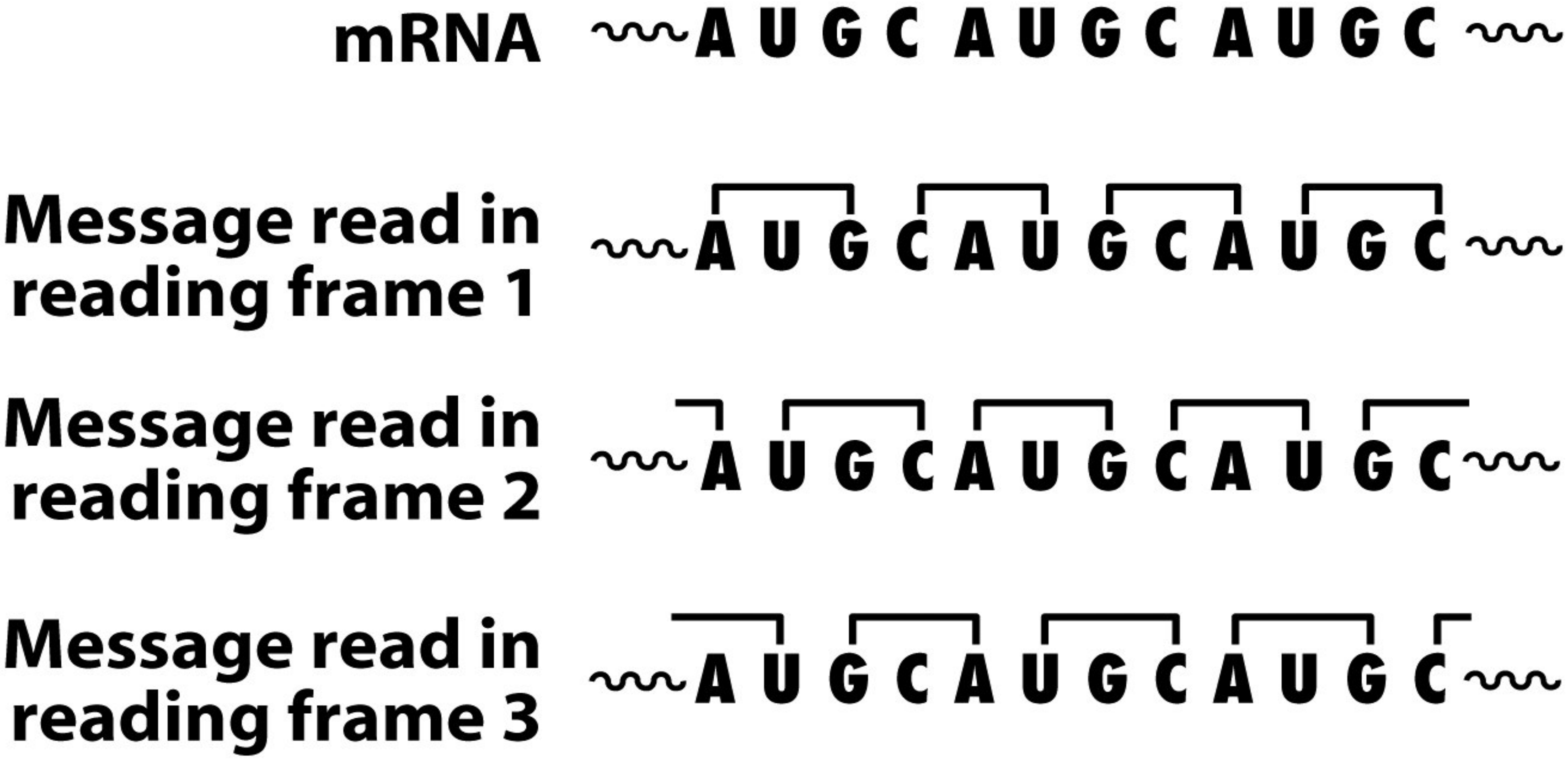
Reading Frame



- A sequence of nucleotides in mRNA is read in sequential sets of three nucleotides which are translated into amino acids.
- There are three possible reading frames in protein synthesis.
- The same mRNA sequence can specify three completely different amino-acid sequences, depending on the 'reading frame‘.

Adapted from: http://www.nature.com/nature/journal/v414/n6865/

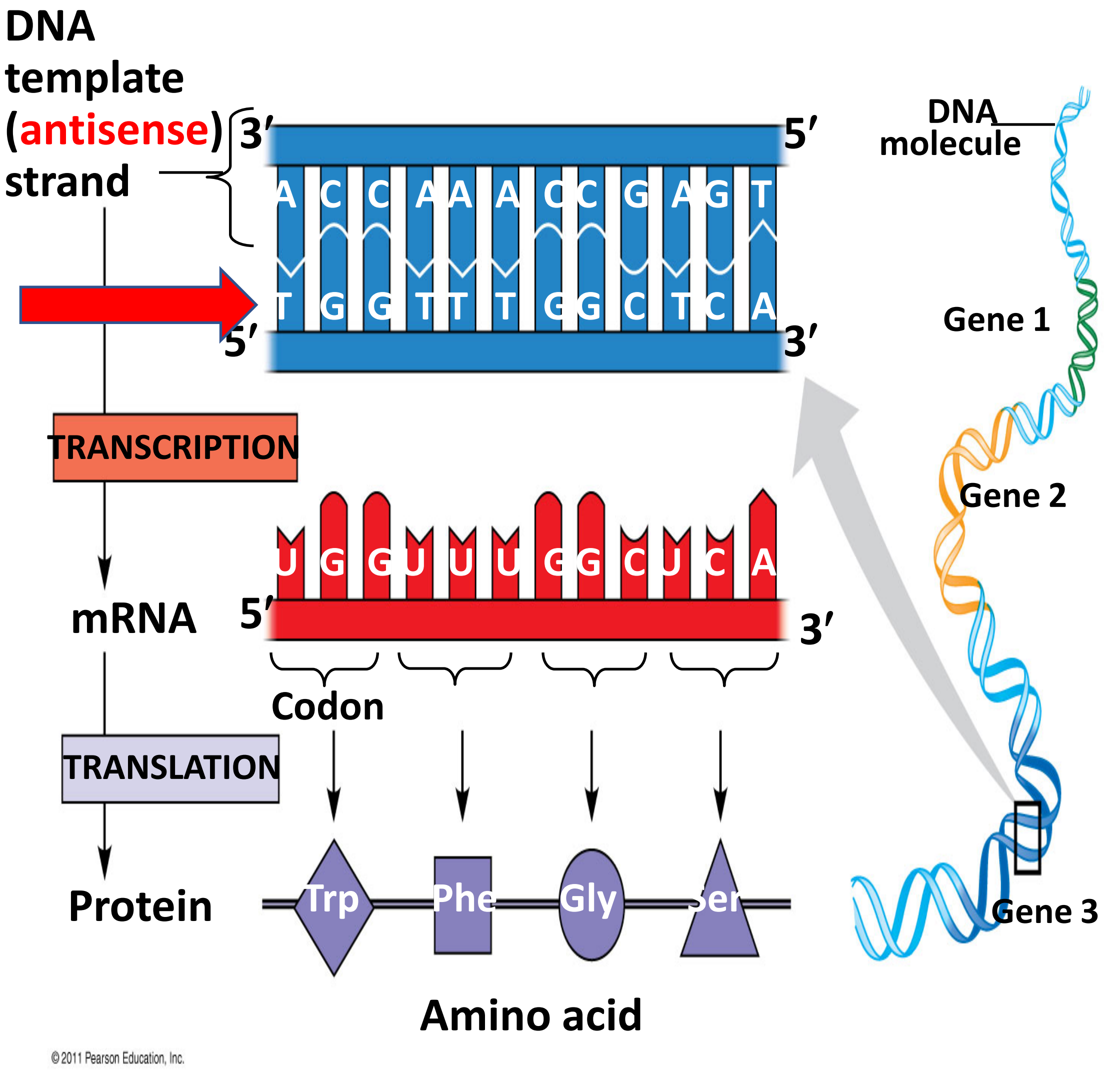
One mRNA contains three reading frames



How many reading frames does dsDNA have?

Label the 5' and the 3' ends of this mRNA.

The Triplet Code & Translation

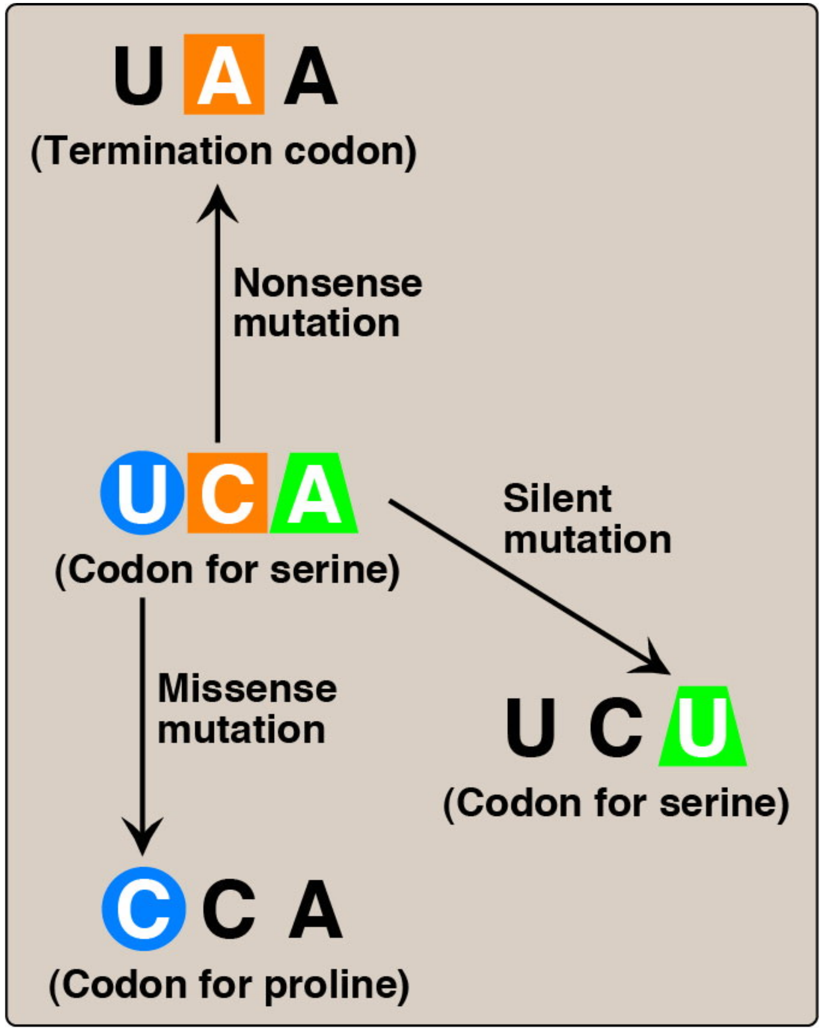


- The template strand is always the same strand for a given gene
- During translation, the mRNA base triplets, called **codons**, are read by translation machinery in the 5′ to 3′ direction
- Each codon specifies the amino acid (one of 20) to be placed at the corresponding position along a polypeptide.

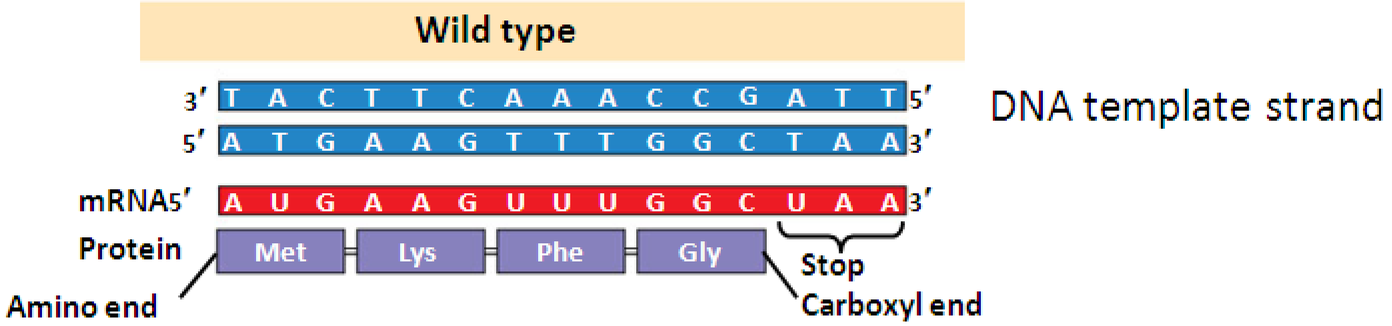
Which is the DNA sense (coding) strand?

Consequences of Altering the Nucleotide Sequence

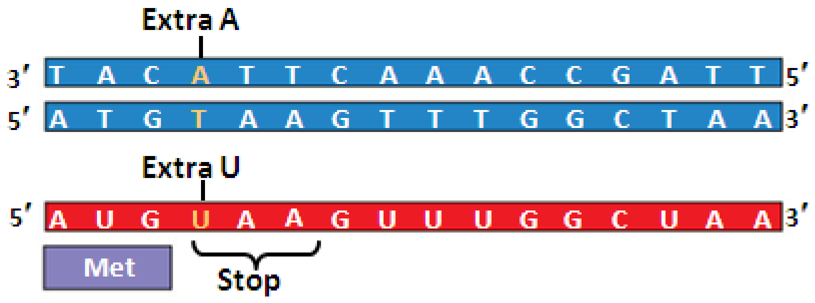
Point Mutation



The Frame-shift Mutation

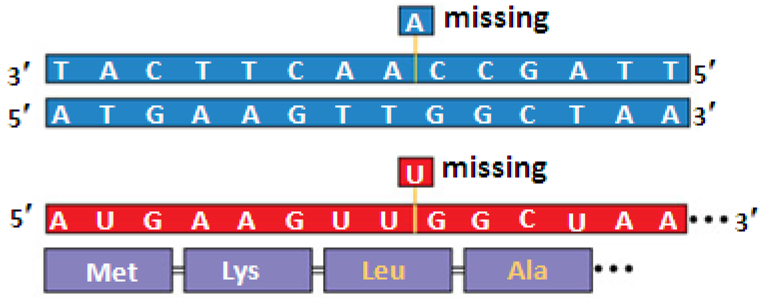


- Insertion or deletion of one base pair
    - Causes a frameshift in the RNA code



Frameshift causing immediate nonsense
(1 nucleotide-pair insertion)

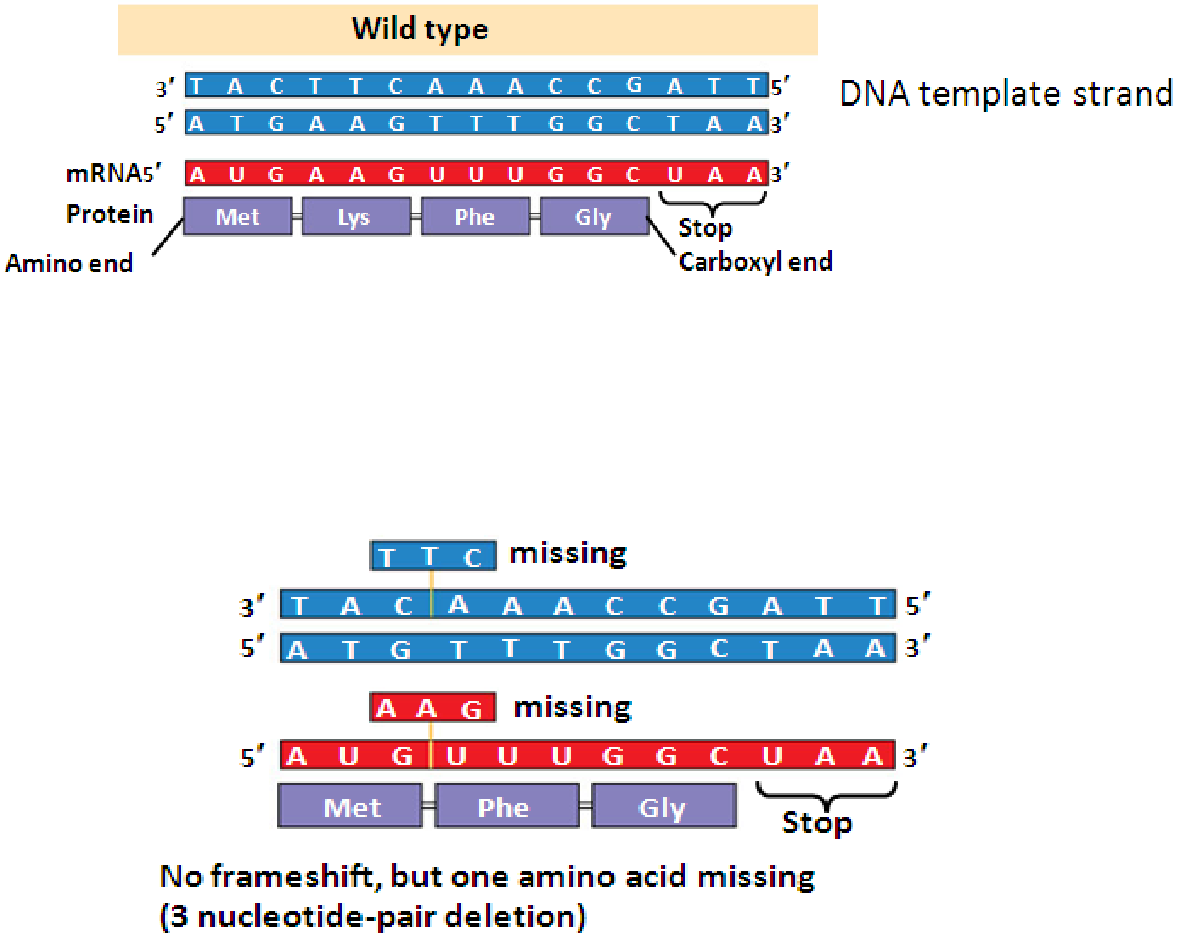
- Insertion or deletion of two base pairs
    - Causes a frameshift in the RNA code



Frameshift causing extensive missense
(1 nucleotide-pair deletion)

The Frame-shift Mutation

- Insertion or deletion of three base pairs
  - Causes a change in amino acid content of protein
  - Insertion or deletion of an amino acid
  - Usually has no effect, but sometimes it does (a classic disease example is the ΔF508 mutation of cystic fibrosis)
- Insertion or deletion of any multiple of 3 nucleotides will lead to larger inframe changes



Thank you.

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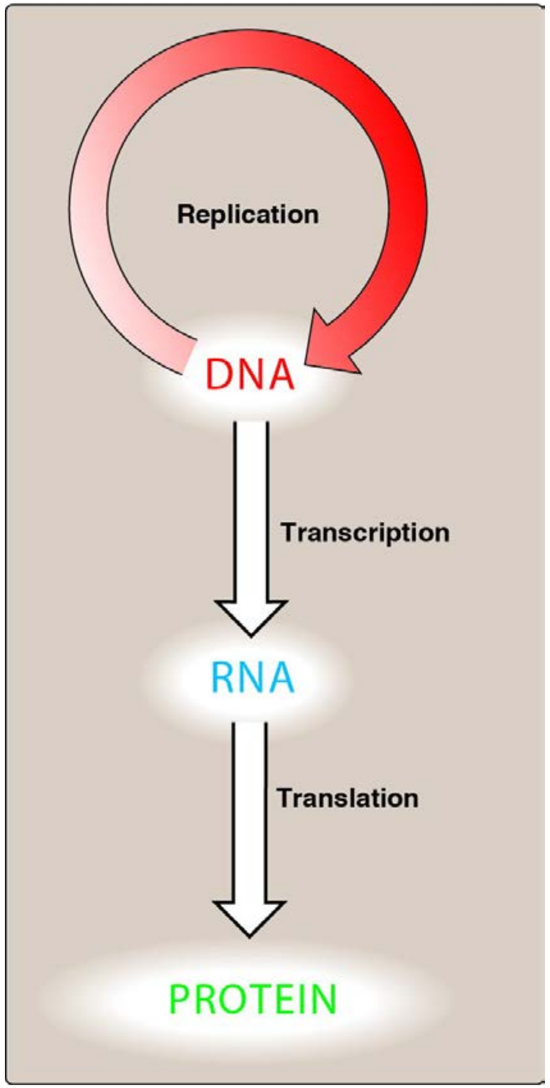
The major RNA species in animal cells

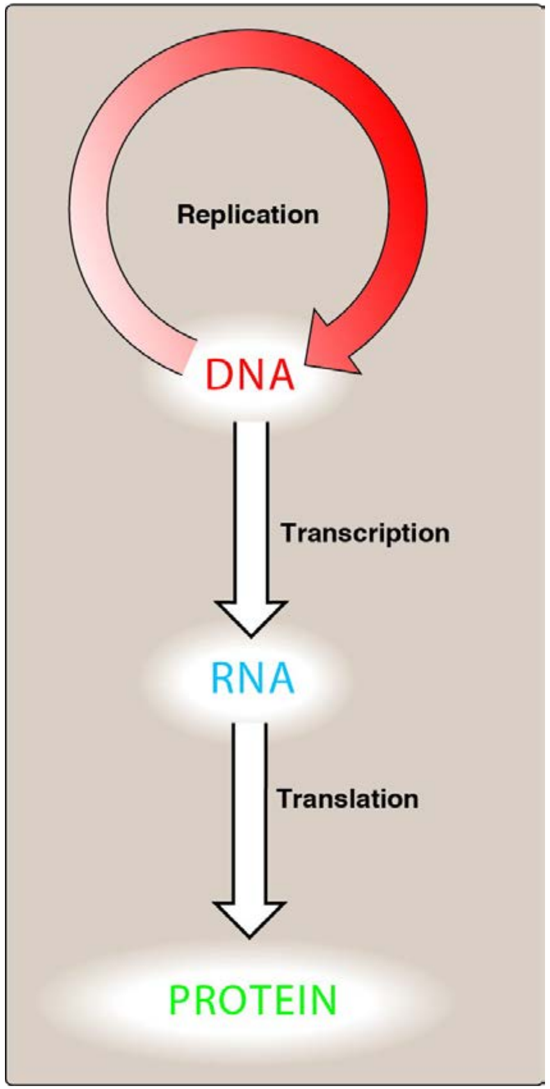
~99.9 %

- rRNA
  - *Ribosomal*
- tRNA
  - *Transfer*
- mRNA
  - *Messenger*
- snRNA
  - *Small nuclear* (splicing of mRNA)
- snoRNA
  - *Small nucleolar*RNA (splicing of rRNA)
- miRNA
  - *MicroRNA* (gene regulation)



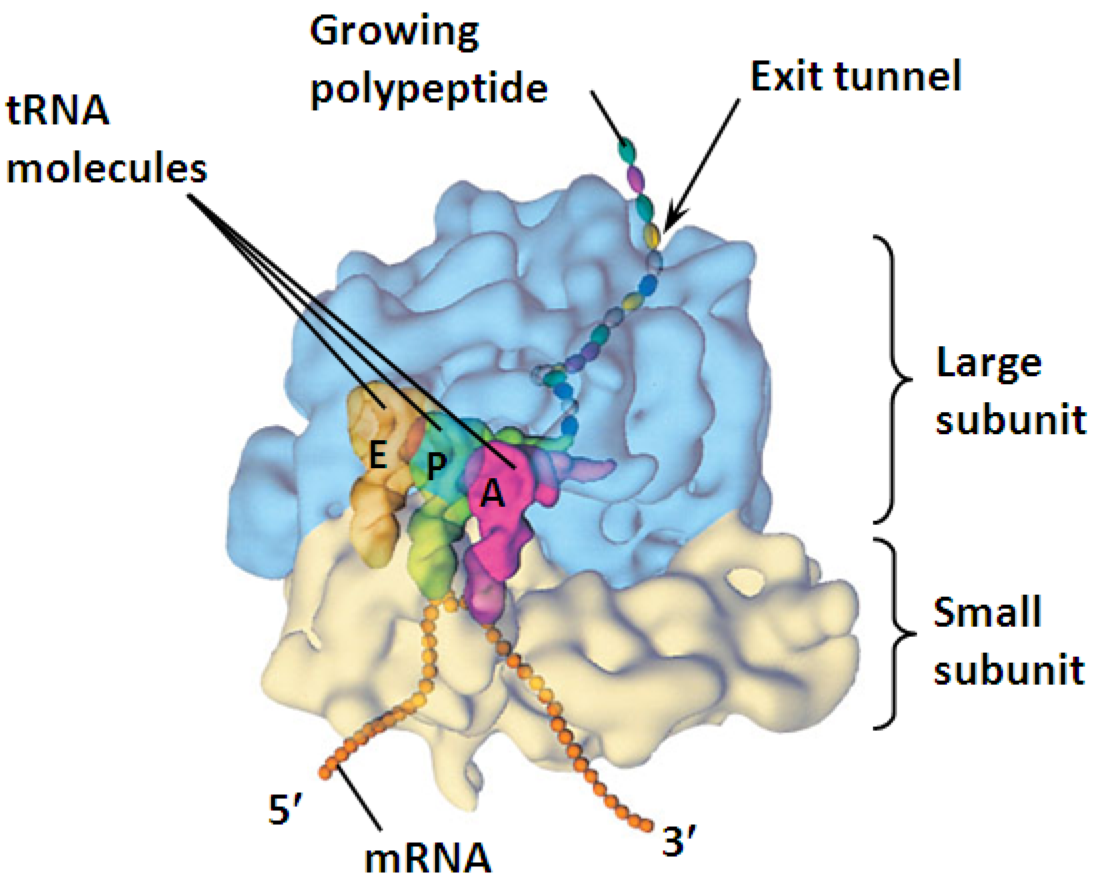
Components Required for Translation

- mRNA
- Ribosomes
- Charged tRNA
- GTP
- Initiation factors



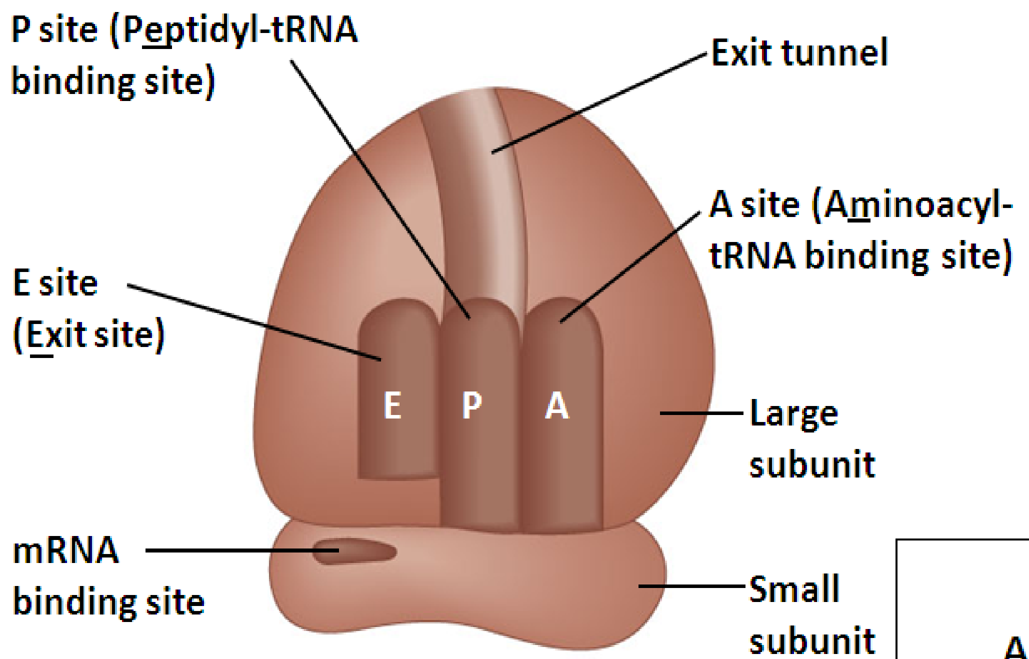
The Anatomy of a Functional Ribosome.

- A ribosome has three binding sites for tRNA
  - The **P site** holds the tRNA that carries the growing polypeptide chain
  - The **A site** holds the tRNA that carries the next amino acid to be added to the chain
  - The **E site** is the exit site, where discharged tRNAs leave the ribosome

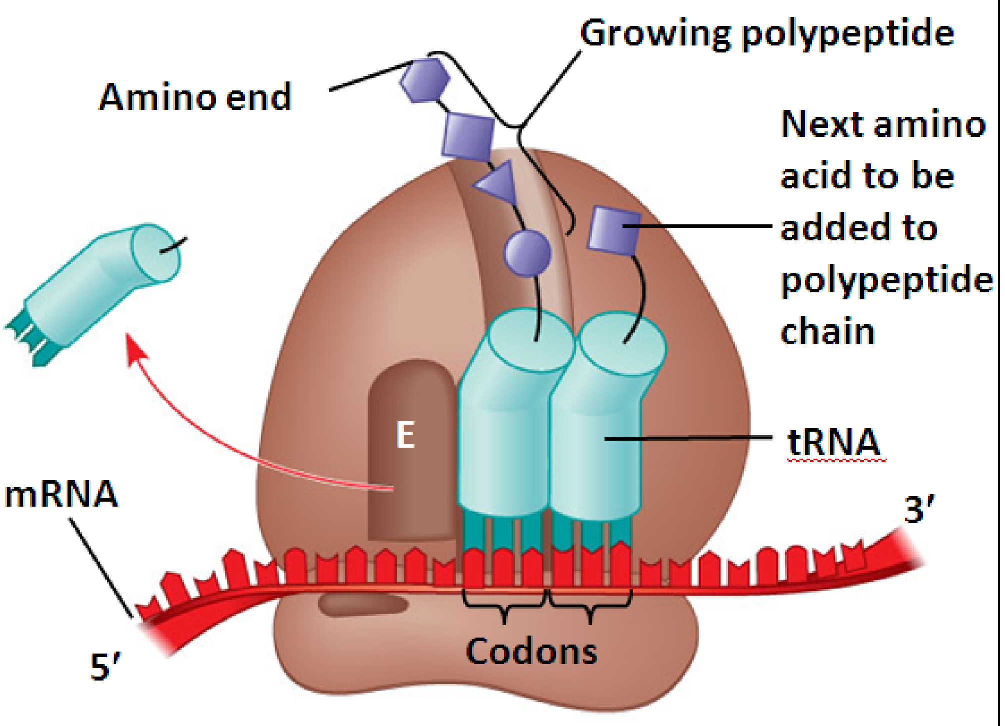


(a) Computer model of functioning ribosome

Schematic Models of a Functional Ribosome



(b) Schematic model showing binding sites



(c) Schematic model with mRNA and tRNA

SOM.MK.I.BPM1.1.FTM.3.BCHM.0070

Eukaryotic and Prokaryotic Ribosomes

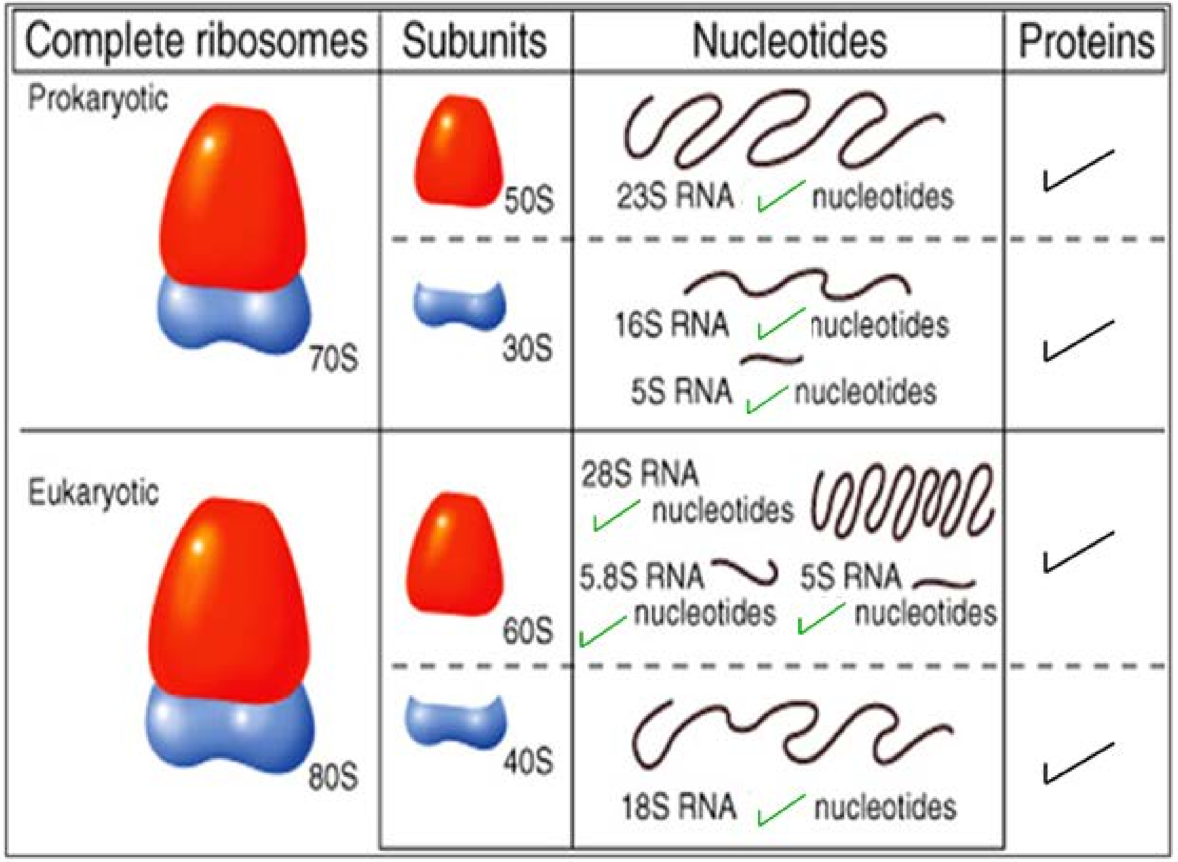
1) The ribosome has two subunits each composed of rRNA and numerous proteins.

2) About 2/3 of the mass of the ribosome is RNA

3) In *E.coli* the small 30S subunit is involved in the initial binding to the mRNA.

4) The larger 50S subunit contributes the peptidyl transferase activity.

5) Both the 30S and 50S contribute to the E-, P- and A- sites.



Adapted from The McGraw-Hills companies, Inc.

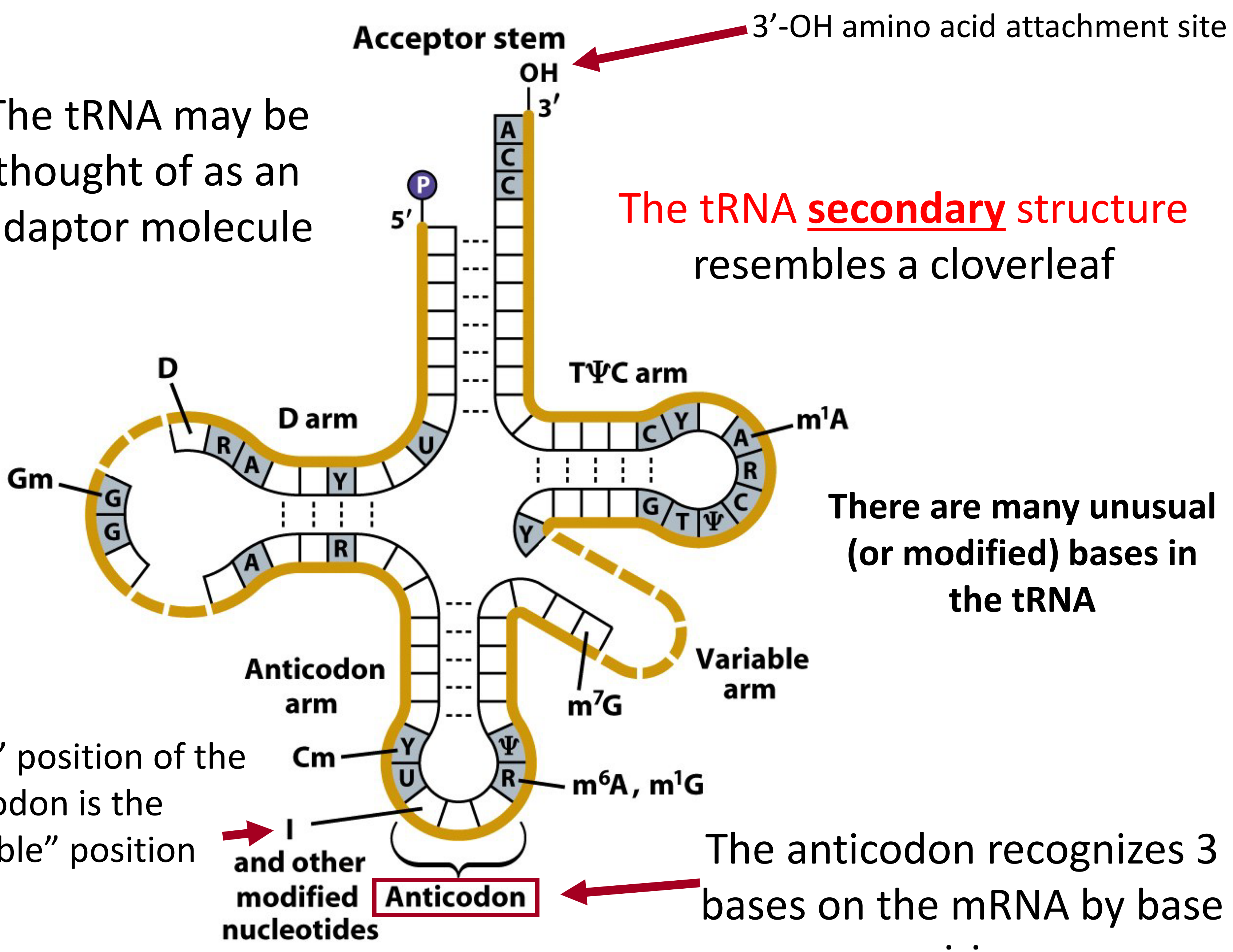
= numerous

The eukaryotic ribosome is bigger and more complex then the prokaryotic

The tRNA may be thought of as an adaptor molecule

The tRNA **secondary** structure resembles a cloverleaf

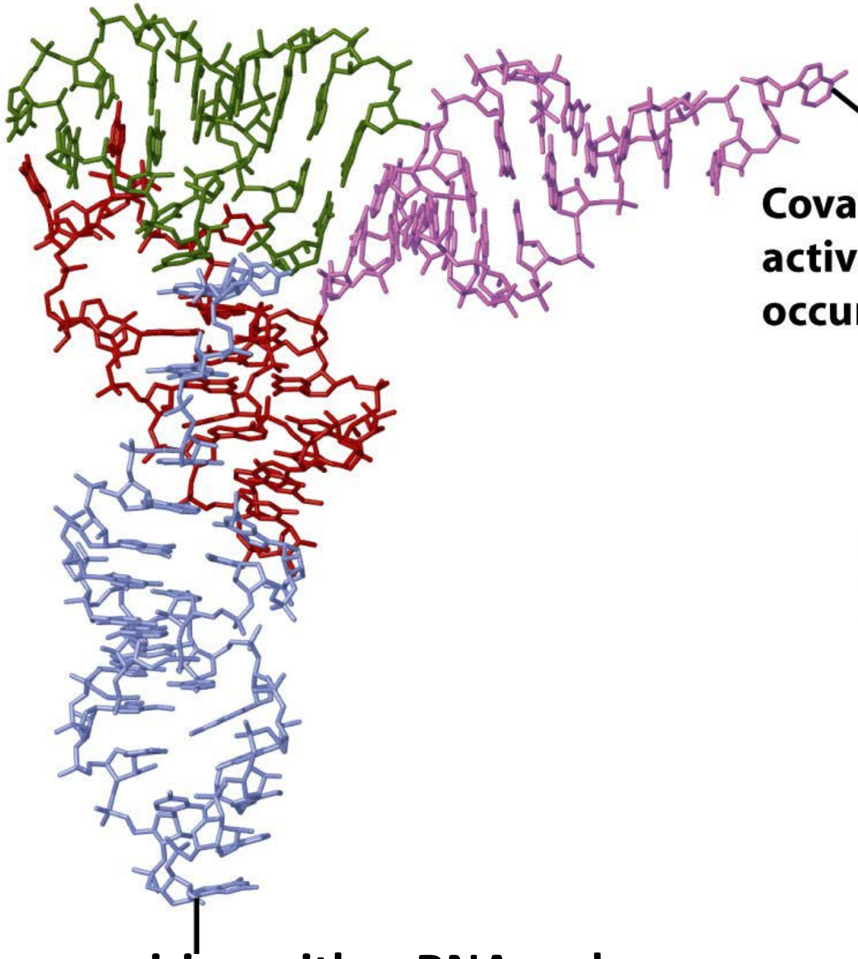
There are many unusual (or modified) bases in the tRNA

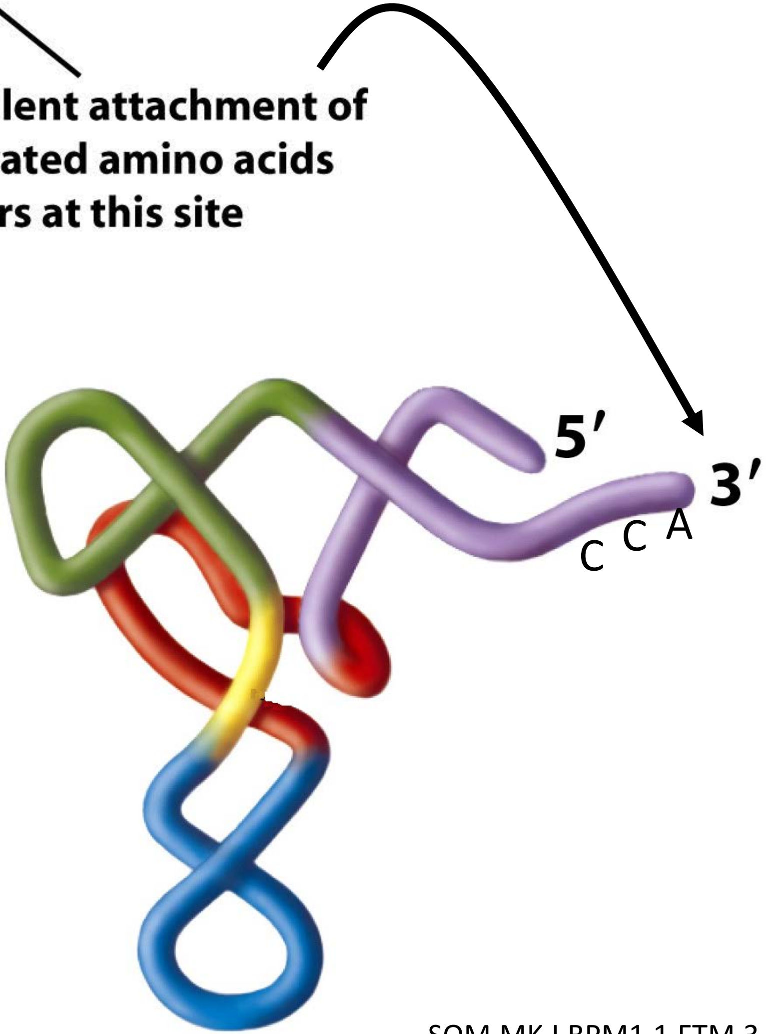


The 5' position of the anticodon is the "wobble" position

The anticodon recognizes 3 bases on the mRNA by base pairing

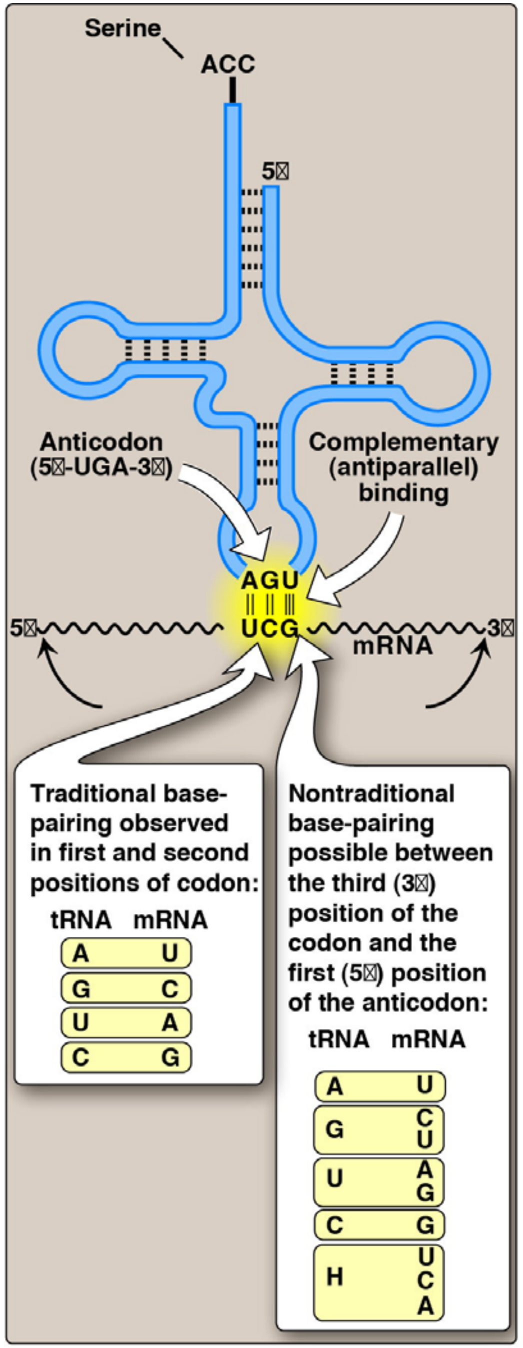
The Tertiary Structure of tRNA is more complex





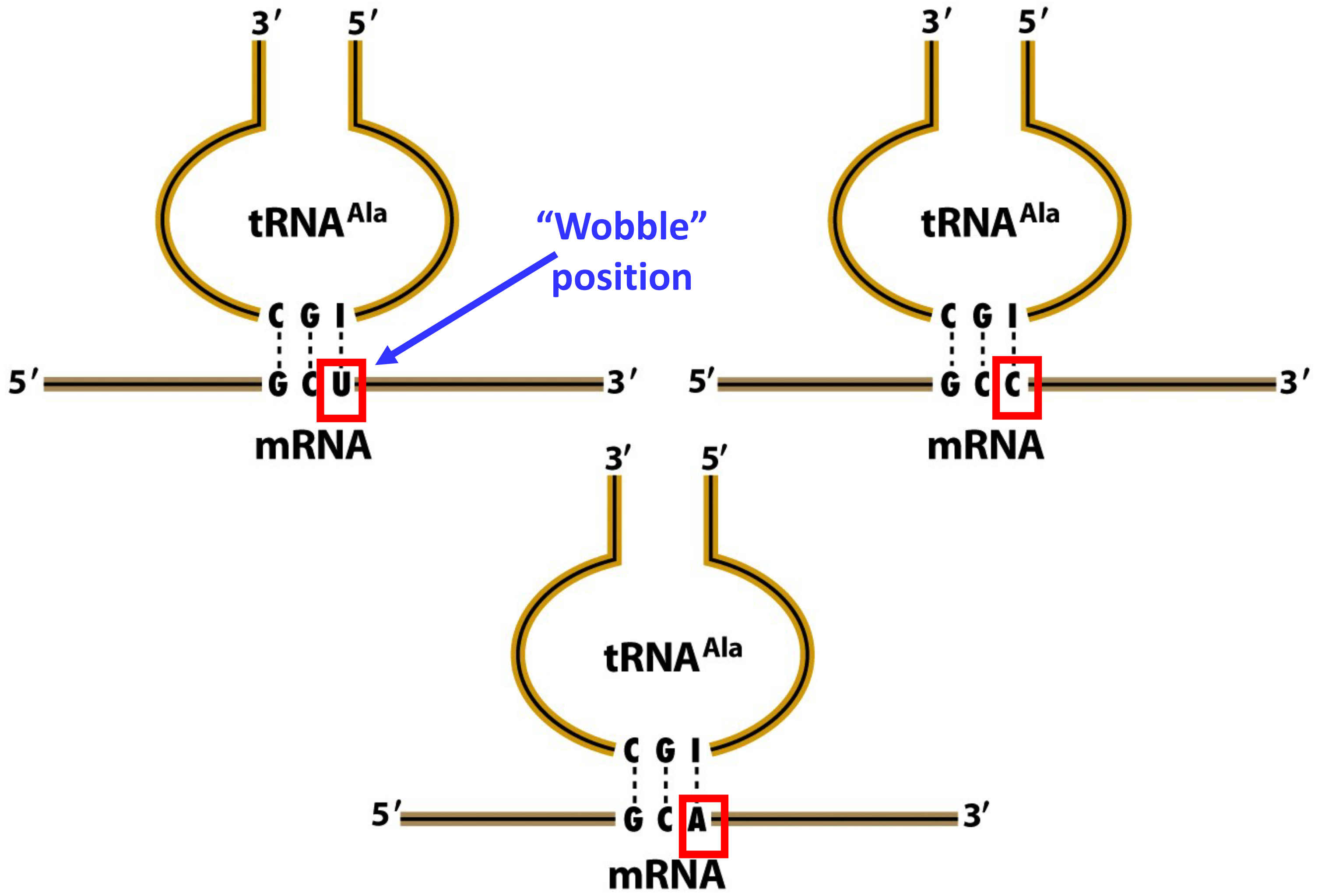
The Wobble Hypothesis

- Codon-anticodon pairing follows traditional rules for first two bases of codon, but less stringent for the last base
- **Wobble:** Nontraditional base pairing between the 5'-nucleotide (first nucleotide) of the anticodon with the 3'-nucleotide (last nucleotide) of the codon.
- **So, less than 61 tRNAs exist, since some tRNAs may read more then one codon**
- **Inosine** (I) is a nucleoside formed when hypoxanthine (base) is attached to a ribose ring.



One tRNA^{Ala} with the IGC anticodon can bind three codons specifying Alanine since I can pair with U, C, or A

- Some amino acids have more than 1 codon.
- Some tRNAs may read more than 1 codon.
- Note the antiparallel arrangement between mRNA and tRNA.



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